

Generation Interconnection System Impact Study Report

For Queue Project Q14-0384

Stepstone 100 kV
November 24, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0384. The Interconnection Customer proposes a 467.4 MW solar photovoltaic facility interconnecting at the Stepstone 100 kV substation in the Indiana area. The request was received on September 22, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 231 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 22 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$27,546,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Sepstone to Air Liquide Substation 100 KV line	Whitestone to Sepstone Substation 100 KV line	421.2	108	100.0
Silver Saddle to Kemper County 100 KV	Whitestone to Stepstone 100 KV	389.2	112	100.0
Whitestone to Stepstone 100 KV	Stepstone to Air Liquide 100 KV	385.0	118	100.0
Air Liuide Substation to Silver addle 100 KV	Stepstone to Whitestone 100 KV	377.8	115	100.0
Whitestone to Molino 100 KV	Stepstone to Air Liquide 100 KV line	347.4	126	100.0
Freehaven-Lakepark to Molino 100 KV line	Stepstone to Air Liquide 100 KV	284.7	141	100.0
High Falls - Croghan Ny to S end 100 KV	Dalreed to Kemper County 100 KV	177.2	154	28.7
Kemper Couny to S Bend 100 KV Line	Keper County to Dalreed Substation 100 KV	159.8	187	28.7
Kemper County to Twin Branch 100 KV	Whitestone to Stestone 100 KV Line	132.1	162	43.0
Suter Substation to High Falls - Croghan Ny 100 KV	Kemper County to Dalreed 100 KV Line	124.2	233	17.7
Suter to Table Mt 100 KV	Kemper County to Dalreed 100 KV Line	123.9	217	17.7
Tortoise to Teeland 100 KV Line 1	Tortoise Substation to Teeland 100 KV Line	121.1	160	17.5
Teeland to Tortoise 100 KV Circuit 2	Tortoise to Teeland Substation 100 KV Line	121.1	160	17.5
Shawville Sta 500/345 kV XFMR	Carbon Center 500/345kV XFMR	108.1	860	5.1
Teeland to Table Mt 100 KV Cir. 1	Table Mt to Teeland 100 KV Line 2	107.5	204	18.3

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Table Mt Substation to Teeland 100 KV Ckt 2	Teeland to Table Mt 100 KV	107.5	204	18.3
Tortoise to Denver Terminal 100 KV L	Denver Terminal to Tortoise 100 KV	106.3	165	17.3
Denver Terminal to Tortoise 100 KV	Tortoise to Denver Terminal 100 KV	106.3	165	17.3
Frehaven-Lakepark to Twin Branch 100 KV	Stepstone to Whitestone Substation 100 KV Line	106.1	179	43.0
Shawville Sta to Blackrock 345 KV	Carbon Ctr 500/345kV Auto	105.9	1,325	7.9
Talenenergy Martins Creek Llc Williamsport to Montour Station 345 KV	Fort Humbug to Orchard Road 500 KV Line	105.5	1,324	5.5
Hilltop 500/345 kV TX	Hilltop to Talenenergy Martins Creek Llc Williamsport 500 KV	103.6	1,694	7.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Air Liquide to Stepstone 100 KV	Rebuild line with higher-capacity conductor	\$1,751,000	100% (\$1,751,000)
Kemper County to Silver Saddle 100 KV L	Rebuild line with higher-capacity conductor	\$533,000	100% (\$533,000)
Whitestone to Stepstone 100 KV Line	Rebuild line with higher-capacity conductor	\$2,911,000	100% (\$2,911,000)
Air Liuide Substation to Silver addle 100 KV	Rebuild line with higher-capacity conductor	\$5,405,000	100% (\$5,405,000)
Whitestone to Molino 100 KV	Rebuild line with higher-capacity conductor	\$3,722,000	100% (\$3,722,000)
Freehaven-Lakepark to Molino 100 KV line	Rebuild line with higher-capacity conductor	\$1,287,000	100% (\$1,287,000)
S Bend to High Falls - Croghan Ny 100 KV Line	Rebuild line with higher-capacity conductor	\$2,762,000	43% (\$1,194,000)
Kemper Couny to S Bend 100 KV Line	Rebuild line with higher-capacity conductor	\$1,953,000	43% (\$844,000)
Kemper County to Twin Branch 100 KV L	Rebuild line with higher-capacity conductor	\$2,270,000	50% (\$1,147,000)
Suter Substation to High Falls - Croghan Ny 100 KV	Rebuild line with higher-capacity conductor	\$1,785,000	26% (\$464,000)
Suter to Table Mt 100 KV	Rebuild line with higher-capacity conductor	\$3,803,000	26% (\$989,000)
Teeland to Tortoise 100 KV Circuit 1	Rebuild line with higher-capacity conductor	\$1,677,000	29% (\$484,000)
Tortoise to Teeland 100 KV	Rebuild line with higher-capacity conductor	\$2,240,000	29% (\$647,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Shawville Sta 500/345kV Auto	Replace transformer with higher-rated unit	\$32,744,000	6% (\$1,871,000)
Teeland to Table Mt 100 KV Line	Reconductor line	\$513,000	28% (\$141,000)
Table Mt Substation to Teeland 100 KV Ckt 2	Reconductor line	\$444,000	28% (\$122,000)
Denver Terminal to Tortoise 100 KV #1	Reconductor line	\$612,000	29% (\$177,000)
Denver Terminal to Tortoise 100 KV	Reconductor line	\$704,000	29% (\$203,000)
Twin Branch to Freehaven-Lakepark 100 KV	Reconductor line	\$677,000	50% (\$342,000)
Shawville Sta to Blackrock 345 KV	Reconductor line	\$16,842,000	4% (\$699,000)
Montour Sta to Talenenergy Martins Creek LLC Williamsport 345 KV	Reconductor line	\$20,837,000	4% (\$865,000)
Hilltop 500/345 kV TX	Replace transformer with higher-rated unit	\$42,105,000	4% (\$1,748,000)
	Total allocated to project		\$27,546,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.